2a

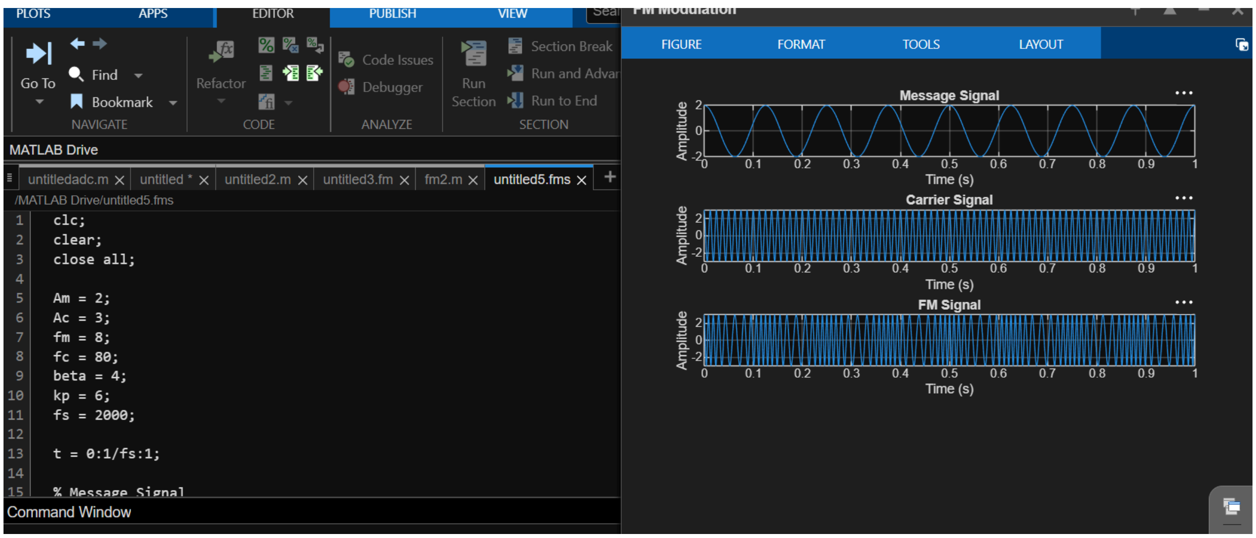
Objectives
(i) To generate a Frequency Modulated (FM) signal using MATLAB by varying the frequency of a high-frequency carrier in accordance with the amplitude of the message signal and to observe the resulting waveform in the time domain.

Program

```
clc;
clear;
close all;

Am = 2;
Ac = 3;
fm = 8;
fc = 80;
beta = 4;
kp = 6;
fs = 2000;

t = 0:1/fs:1;

% Message Signal
m = Am * cos(2*pi*fm*t);

% Carrier Signal
c = Ac * cos(2*pi*fc*t);

% FM Signal
fm_sig = Ac * cos(2*pi*fc*t + beta*sin(2*pi*fm*t));

% Plot Signals
figure('Name','FM Modulation','NumberTitle','off');

subplot(4,1,1);
plot(t,m);
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(4,1,2);
plot(t,c);
```

```
title('Carrier Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(4,1,3);
plot(t,fm_sig);
title('FM Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
```

Output



Result

The Frequency Modulated (FM) signal was successfully generated using MATLAB, and the time-domain waveforms of the message signal, carrier signal, and FM signal were observed and analyzed.

2b

Objective

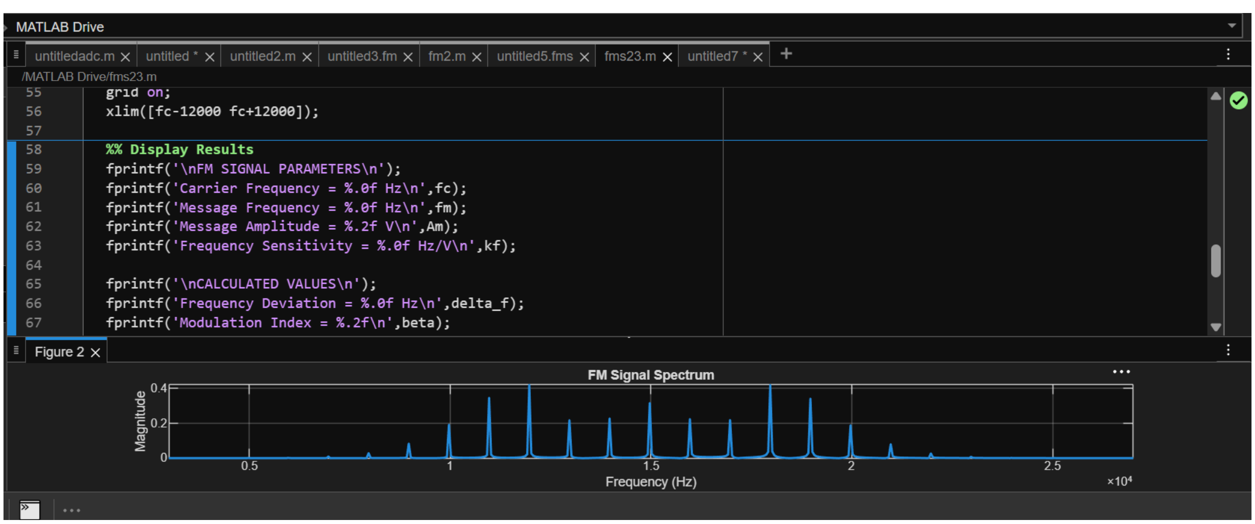
(ii) To analyze the frequency spectrum of the generated FM signal using FFT and identify the carrier component, sideband frequencies, and the bandwidth of the FM signal based on Carson's rule.

Program

```
clc;
clear;
close all;

%% Input Parameters
Am = 3;
fm = 1000;
Ac = 2;
fc = 15000;
kf = 1500;
fs = 200000;
T = 0.02;

t = 0:1/fs:T;

%% Message Signal
m = Am*cos(2*pi*fm*t);

%% Frequency Deviation
delta_f = kf*Am;

%% Modulation Index
beta = delta_f/fm;

%% FM Signal
fm_signal = Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t));

%% Message Signal Plot
figure;
subplot(2,1,1);
plot(t,m,'LineWidth',1.5);
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

%% FM Signal Plot
subplot(2,1,2);
plot(t,fm_signal,'LineWidth',1.5);
title('Frequency Modulated Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
```

```
%% Spectrum Analysis
N = length(fm_signal);
FM_fft = fftshift(abs(fft(fm_signal))/N);
f = (-N/2:N/2-1)*(fs/N);

figure;
plot(f,FM_fft,'LineWidth',1.5);
title('FM Signal Spectrum');
xlabel('Frequency (Hz)');
ylabel('Magnitude');
grid on;
xlim([fc-12000 fc+12000]);

%% Display Results
fprintf('\nFM SIGNAL PARAMETERS\n');
fprintf('Carrier Frequency = %.0f Hz\n',fc);
fprintf('Message Frequency = %.0f Hz\n',fm);
fprintf('Message Amplitude = %.2f V\n',Am);
fprintf('Frequency Sensitivity = %.0f Hz/V\n',kf);

fprintf('\nCALCULATED VALUES\n');
fprintf('Frequency Deviation = %.0f Hz\n',delta_f);
fprintf('Modulation Index = %.2f\n',beta);

BW = 2*(delta_f+fm);
fprintf('Bandwidth (Carson Rule) = %.0f Hz\n',BW);
```

Output



Result
i) The time-domain waveform was observed, and the frequency spectrum obtained using FFT has
been plotted.
ii) Also, calculated the frequency deviation and bandwidth using Carson’s rule.